

Chapter 9: Matrices and Matrix Arithmetic - Summary

Chapter 9 introduces matrices and matrix arithmetic. Key concepts:

- A matrix is a two-dimensional array of numbers.
- Matrices are used to represent datasets, transformations, and systems of equations.

Operations covered:

1. Matrix addition
2. Matrix subtraction
3. Hadamard (element-wise) multiplication
4. Matrix division (element-wise)
5. Matrix-matrix multiplication (dot product)
6. Matrix-vector multiplication
7. Matrix-scalar multiplication

Most important concept: Matrix multiplication. If A is $m \times n$ and B is $n \times k$, then: $C = A @ B$. The inner dimensions must match. The resulting matrix has dimensions $m \times k$.

Machine-learning connection: Matrix multiplication is the mathematical engine behind:

- Linear regression
- Neural networks
- Dimensionality reduction
- State estimation
- Deep learning